

HDOC Day-12 Activity

Question:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Find the sum and product of the first and last digits.
2. Count how many digits are lesser than, equal to, and greater than a given digit k.

Step 1: Understanding the Problem

Input: A positive integer n, menu choice, and digit k for option 2.

Output: For choice 1, sum and product of first and last digits. For choice 2, counts of digits lesser than, equal to, and greater than k.

Step 2: Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	n=12345, choice=1	Sum=6, Product=5	Sum=6, Product=5	Pass
2	n=9072, choice=1	Sum=11, Product=18	Sum=11, Product=18	Pass
3	n=12345, choice=2, k=3	Lesser=2, Equal=1, Greater=2	Lesser=2, Equal=1, Greater=2	Pass
4	n=5555, choice=2, k=5	Lesser=0, Equal=4, Greater=0	Lesser=0, Equal=4, Greater=0	Pass
5	n=86420, choice=2, k=4	Lesser=2, Equal=1, Greater=2	Lesser=2, Equal=1, Greater=2	Pass

Step 3: Algorithm and Evaluation

1. Start.
2. Read a positive integer n.
3. Display menu: 1 - Sum and product of first and last digits; 2 - Compare digits with k.
4. Read choice.
5. Use switch(choice).
6. For choice 1:
 - a. last = n % 10.
 - b. Repeatedly divide a copy of n by 10 until it becomes a single digit; this is first.
 - c. sum = first + last; product = first * last.
 - d. Display sum and product.
7. For choice 2:
 - a. Read digit k.
 - b. Set lesser = equal = greater = 0.
 - c. Extract each digit using % 10 and compare it with k.
 - d. Increment the appropriate counter and remove the digit using / 10.
 - e. Display all three counts.
8. For any other choice, display Invalid choice.
9. Stop.

Evaluation: The algorithm uses switch-case for menu selection and digit extraction with modulo and integer division. It correctly produces the required results for the prepared test cases.

Step 4: C Programming

```
#include <stdio.h>

int main()
{
    int n, choice, temp, first, last, sum, product;
    int k, digit, lesser = 0, equal = 0, greater = 0;

    printf("Enter a positive integer: ");
```

```

scanf("%d", &n);

printf("1. Sum and product of first and last digits\n");
printf("2. Count digits lesser, equal and greater than k\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        last = n % 10;
        temp = n;
        while (temp >= 10)
            temp = temp / 10;
        first = temp;
        sum = first + last;
        product = first * last;
        printf("Sum = %d\n", sum);
        printf("Product = %d\n", product);
        break;

    case 2:
        printf("Enter digit k: ");
        scanf("%d", &k);
        temp = n;
        while (temp > 0)
        {
            digit = temp % 10;
            if (digit < k) lesser++;
            else if (digit == k) equal++;
            else greater++;
            temp = temp / 10;
        }
        printf("Count_lesser = %d\n", lesser);
        printf("Count_equal = %d\n", equal);
        printf("Count_greater = %d\n", greater);
        break;

    default:
        printf("Invalid choice\n");
}
return 0;
}

```

Step 5: Compilation and Execution

The program was compiled and executed in Kali Linux and evaluated against the prepared test cases.

Step 6: Kali Linux Compilation and Execution

Step 6: Kali Linux Compilation and Execution

The following terminal-style screenshots show compilation and execution for the prepared test cases.

```
■■■(kali■kali)-[~]
■■$ nano day12.c
■■■(kali■kali)-[~]
■■$ gcc day12.c -o day12
■■■(kali■kali)-[~]
■■$ ./day12
Enter a positive integer: 12345
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 1
Sum = 6
Product = 5
```

Test Case 2

```
■■■(kali■kali)-[~]
■■$ ./day12
Enter a positive integer: 9072
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 1
Sum = 11
Product = 18
```

Test Case 3

```
■■■(kali■kali)-[~]
■■$ ./day12
Enter a positive integer: 12345
1. Sum and product of first and last digits
2. Count digits lesser, equal and greater than k
Enter your choice: 2
Enter digit k: 3
Count_lesser = 2
Count_equal = 1
```

```
Count_greater = 2
```